# Homework 5

BIOSTAT 740, Fall 2026

Due October 23, 8:00 PM

Purpose:

1. See how the mixed effects model works its magic
2. Fit a mixed model and evaluate the results

## HW 5

Hint: Parts 1, 2, and 3 will take 5 minutes in R

1. If $Z_1 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 10 \end{pmatrix}$ and $G = \begin{pmatrix} 1 & .1 \\ .1 & 1 \end{pmatrix}$ and $\sigma = 1$, what are $cov(Y_1), corr(Y_1)$ ?

2. If $Z_2 = \begin{pmatrix} 1 & 1.8 \\ 1 & 2 \\ 1 & 4 \\ 1 & 4.2 \end{pmatrix}$ and $G$ and $\sigma$ are as above, what is $cov(Y_2), corr(Y_2)$ ?

3. Compare and contrast the correlation between visit 1 and visit 4 for these two subjects, and the correlation between visits 2 and 3.

4. Use the Exercise data, with times 0, 4, 6, 8, and 12. In *both* SAS (proc mixed) and R (lme4 library, use function `lmer()`) fit a model with a fixed intercept and a fixed linear slope for each intervention group as well as a random intercept and slope for each person.

    a. Is there a difference in how fast strength is built?
    b. How quickly is strength built in each group?
    c. What's the variance of the random slopes around the mean slope?
    d. What's the variance of the random intercepts around the mean intercept?
    e. What's the covariance of the random intercepts and slopes?
    f. What's the correlation between the random intercepts and slopes?